



Symposium review: The role of adipose tissue in transition dairy cows: Current knowledge and future opportunities*

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ABSTRACT

Adipose tissue (AT) is a central reservoir of energy stored in the form of lipids. In addition, AT has been recognized as an immunologically and endocrinologically active tissue of dairy cattle. The recent literature on AT biology of transition dairy cows has often focused on the possible negative effects that originate from excessive body fat. However, the highly efficient energy-storage capability of this tissue is also vital to the adaptability of dairy cattle to the change in nutrient availability, and to support lactation and reproduction. An excessive degree of mobilization of this tissue, however, is associated with high circulating fatty acid concentrations, and this may have direct and indirect negative effects on reproductive health, productivity, and disease risk. Furthermore, rapid lipolysis may be associated with postpartum inflammation. Research on the role of AT is complicated by the greater difficulty of accessing and measuring visceral AT compared with subcutaneous AT. The objective of this review is to provide a transition cow-centric summary of AT biology with a focus on reviewing methods of measuring AT mass as well as to describe the importance for production, health, and reproductive success.

Key words: adipose tissue, body condition, transition period, health

INTRODUCTION

Adipose tissue (AT), specifically white AT, is the major energy reserve in the form of triacylglycerol (TAG; an ester of glycerol and 3 fatty acids) in mammals (Ahmadian et al., 2010). When energy intake is in excess of maintenance and other needs, AT accre-

tion occurs. Brown adipose tissue serves primarily as a source of heat and, in contrast to humans (Cohen and Spiegelman, 2016), has not been detected in cattle beyond the neonatal period (Smith and Carstens, 2005). Uncertainty exists if this is due to the lack of advanced methods such as positron emission tomography or computed tomography phenotyping and other imaging approaches (Chondronikola et al., 2018).

Although an increase in lipolysis is already present prepartum, parturition and the immediate postpartum period are characterized by an increased rate of hydrolysis of fatty acids (FA) from AT TAG (McCarthy et al., 2015). For a detailed synopsis of the physiological and molecular processes and mediators of lipolysis and lipogenesis, the reader is directed to the existing excellent reviews on the topic (Roche et al., 2009; Contreras et al., 2017; Zachut and Contreras, 2022).

In circulation, concentrations of FA (also referred to as nonesterified fatty acids; NEFA) are interpreted as increased rate of lipolysis, decreased peripheral uptake, decreased re-esterification, or a combination of these (Bell, 1995). Magnitude and timing of maximal NEFA concentration postpartum varies considerably between individual cows (McCarthy et al., 2015) and in response to feeding (Quiroz-Rocha et al., 2010). Fatty acids are used for β -oxidation and generation of energy intermediates in the TCA cycle that promote hepatic gluconeogenesis, but can also be incompletely oxidized to ketone bodies or re-esterified as TAG (Bell, 1981). In addition, the released glycerol serves as a direct substrate for hepatic gluconeogenesis (Aschenbach et al., 2010). Approximately half of the AT-released NEFA are either oxidized or incorporated into milk triglycerides in the periparturient cow (Bell, 1995). Arterial concentrations of NEFA determine 56% of the variability in individual cow's mammary gland uptake of NEFA, meaning that as NEFA concentrations increase, mammary uptake increases (Miller et al., 1991).

This review focuses on the role of AT in transition cows with particular attention to a review of techniques to approximate internal and external AT mass as well as to describe the importance of this tissue for production, health, and reproductive success.

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APPROACHES TO APPROXIMATING OR MEASURING ADIPOSE TISSUE MASS OF DAIRY COWS

Subcutaneous and visceral AT mass undergoes changes during the transition period (Mann et al., 2016b; Szura et al., 2020; Knob et al., 2021). Any assessment of the effects of AT mobilization or deposition, as well as the ability to measure effects of management interventions, must rely on an accurate method to describe AT mass in the body. Direct measurement of AT mass relies on slaughter studies and carcass composition (Drackley et al., 2014). Indirect measurements of subcutaneous or total AT depots and rate of lipolysis have been developed to approximate these measurements in the live animal. Determination of BCS by visual scoring systems is an inexpensive and rapid indirect method to approximate a cow's subcutaneous AT depots (Edmonson et al., 1989), but differences in muscle mass cannot be separated in the scoring (Siachos et al., 2021). One unit in BCS change on a 5-point scale has been estimated to correspond to 40 to 55 kg of empty body fat (Komaragiri and Erdman, 1997; Komaragiri et al., 1998). Accuracy, consistency, and interconversion among different BCS scales used in dairy systems worldwide have been reviewed (Roche et al., 2009).

Although BCS has been used widely, animals with similar subcutaneous adipose tissue mass and BCS may have close to a 2-fold weight difference of visceral AT, and correlations between BCS and visceral AT depots were only moderate (Drackley et al., 2014). Visceral AT mass also expanded more rapidly in cows overfed energy, and this difference was not reflected in a weight or BCS difference (Drackley et al., 2014). Therefore, visual scoring systems are likely inaccurate methods to describe visceral AT mass. This is also shown in the wide range of the ratio of subcutaneous to total abdominal AT from 0.38 to 0.83 in early-lactation cows (≤ 100 DIM, $n = 6$; Raschka et al., 2016), and from 0.23 to 0.31 in late-pregnant cows (10–13 d before expected parturition, $n = 10$; De Koster et al., 2015). Automated camera-based BCS systems have been developed and evaluated in recent years and have acceptable accuracy in the middle BCS ranges, although error increases in low- and high-BCS cows (Mullins et al., 2019). Because camera systems rely on measurements of outer cow contours in a similar fashion as visual scoring systems, the same limitations as discussed above apply to this method, but the unbiased and automated measurement is desirable.

Transcutaneous ultrasonography has been used to describe both the subcutaneous (Schröder and Staufienbiel, 2006; Mann et al., 2016b) and to estimate the visceral AT mass (Raschka et al., 2016). Ultrasono-

graphically-determined backfat thickness (**BFT**) has been shown to have an acceptably high correlation with BCS (Hussein et al., 2013; Siachos et al., 2021) and a high correlation of the thurl site ($r = 0.9$) with total body fat content (Schröder and Staufienbiel, 2006). Siachos et al. (2021) determined that 1 unit of BCS change on a 5-point BCS scale was associated with 8.2 mm of BFT measured at the thurl site. However, care should be taken in the interpretation of values derived from different studies as ultrasonography techniques can differ greatly between studies, such as in landmarks and whether skin thickness is included in measurements (Schröder and Staufienbiel, 2006; Siachos et al., 2021), or not (Mann et al., 2016b; Bell et al., 2018). In the author's experience, even small changes in probe placement or pressure applied to the interface can lead to large variability in BFT measurements; therefore, repeated measurements on animals over time should rely on marking the exact location, for example, by clipping the site for consistent probe placement (Mann et al., 2016b). Any compression of the tissue must be avoided.

A limitation of current widely applied ultrasonographic measurements is that internal adipose depots are not assessed. Recently, estimation of visceral AT by a series of ultrasound measurements of the retroperitoneal, mesenteric, and omental AT has been described using 23 German Holstein cows for model development and 6 cows for model evaluation (Raschka et al., 2016). The developed approach was used in transition period studies of the same group (Schäfers et al., 2017; Ruda et al., 2019; Szura et al., 2020). Given that this approach is noninvasive and the necessary probes (6 MHz linear, 3 MHz convex) are available for most portable bovine ultrasound machines today, this method has the potential to greatly improve AT depot estimation by ultrasound. Future studies with larger animal numbers could further evaluate which of the measurements may be best suited to adequately describe the rapid changes in visceral adipose depots of transition dairy cows, and which depots are most relevant for metabolic, health, and production outcomes. Method accuracy has not been validated in other independent studies of Holstein cows or cows of different breeds.

Other indirect methods are currently used for estimating AT mass in research. Several different dilution methods using stable (deuterium oxide) or radioactive isotopes ($[^{40}\text{K}]$, tritium oxide), or urea dilution have been described for research purposes; of these, deuterium oxide, a nonmetabolizable, nonradioactive, and easily quantifiable substance, appears most useful for dilution methods to approximate total body fat in cattle (Martin and Ehle, 1986). It can be further improved by including BW and BFT in the calculation (Schäff et al., 2017). Bioelectrical impedance measurements

(effective electrical resistance), a noninvasive method using self-adhesive current and voltage electrodes, was found to more accurately approximate total body AT mass rather than subcutaneous AT alone (Schäff et al., 2017). Authors concluded that this method is superior to the deuterium dilution method, but further validation is still needed to determine the usefulness of this approach on a larger number of cows. (Schäff et al., 2017). In addition, although these initial results for bio-electrical impedance were promising, and the method could be used noninvasively, it is not currently used in the field and has not been applied in longitudinal studies to describe the rapid AT changes (e.g., in the transition period). Other methods that are of interest to research applications include computed tomography as described by Rosenblatt et al. (2017), as well as magnetic resonance imaging and dual-energy X-ray absorptiometry (reviewed by Scholz et al., 2015), but these have not been used to assess body fat in adult dairy cattle to date due to animal size.

THE IMPORTANCE OF TRANSITION PERIOD ADIPOSE TISSUE ON MILK PRODUCTION

Adipose tissue reserves, and in particular the ability to mobilize AT, support the rapidly changing energetic cost of milk production in early lactation. Bauman and Currie (1980) calculated that the energy deficit of cows in the first 10 wk of lactation was equivalent to approximately 50 kg of pure lipid, and mobilized body reserves in the first 4 wk of lactation had an energy-equivalent of one-third of the milk produced in that time. Cows overfed energy prepartum and with greater BFT had consistently higher milk fat in colostrum and milk over the first months of lactation than control cows, a difference explained by greater incorporation of de novo FA into colostrum (Mann et al., 2016a), whereas a numerically higher preformed FA yield was apparent in milk samples in wk 2 and 4 (Mann et al., 2015). These data show that energy deposition in the prepartum period due to overfeeding influences the composition of milk fat in early lactation.

With these numbers in mind, it is not surprising that cows were traditionally overfed in late lactation, and particularly in the dry period, to prepare the energy stores for these lactational needs (Bauman and Currie, 1980), such that cows could “milk the fat off their back” during the time of greatest energy deficit early postpartum. Indeed, cows with greater BFT and BCS have the greatest increase in NEFA concentrations postpartum (Weber et al., 2016; Szura et al., 2020) and mobilize a greater AT mass, as demonstrated by ultrasonography of back fat (van der Drift et al., 2012) and estimation of subcutaneous and abdominal AT (Szura

et al., 2020). Mobilization of AT is facilitated by low circulating glucose concentrations and a concurrent drop in circulating concentrations of insulin (a potent antilipolytic hormone), as well as a growth hormone-facilitated increase in sensitivity to β -adrenergic stimulation of lipolysis (reviewed in Contreras et al., 2017). A drop in circulating insulin concentrations appears to be important to permit lipolysis postpartum because the inhibition of whole-body lipolysis by glucose-induced insulin response is maintained postpartum (Mann et al., 2016d).

When examining effects of AT on production outcomes, current management of dairy cows most often uses visual BCS as a proxy for cow AT mass. Limitations presented above, however, show that cows with equal BCS can have different internal AT depots that may contribute to production effects, particularly when taking into account that visceral AT has a higher lipolytic activity compared with subcutaneous AT (Ruda et al., 2019). The effect of body condition on milk production reported in several studies has been reviewed and summarized by Roche et al. (2009), who proposed a curvilinear relationship of calving BCS and production, meaning that on both ends of the BCS scale, milk production is decreased compared with the optimum of 3.0 to 3.25 based on all assessed parameters. Roche et al. (2015) more recently compared milk production and components in cows with a BCS of 4 versus a BCS of 5 on a 10-point scale (corresponding to 2.78 and 3.10 on a 5-point scale; Roche et al., 2004). Irrespective of close-up feeding strategy also tested in this study, cows with BCS of 5 produced more milk and more milk fat, protein, and lactose than cows with BCS 4 over the first 12 wk of lactation, but lost more body condition and had 53% higher blood NEFA concentrations during that time (Roche et al., 2015), indicating a greater rate of lipolysis. Similarly, Chapinal et al. (2012a) reported that cows with a BCS ≥ 4.0 in the week before calving produced 1.4 kg/d more milk than cows with BCS ≤ 3.0 . These data would support the traditionally assumed benefit of greater AT stores for milk production. Several multiherd studies, however, have clearly made the association between increased rates of lipolysis (expressed as high postpartum NEFA concentration) and lower milk production, at least in multiparous animals. Higher NEFA concentrations postpartum in animals from 91 different herds (2,290 animals total) across the northeastern United States were associated with lower milk yield in cows (≥ 0.72 mEq/L of NEFA; -647 kg of mature-equivalent 305-d yield) but not in heifers (≥ 0.57 mEq/L of NEFA; $+488$ kg of mature-equivalent 305-d yield; Ospina et al., 2010b). Ospina et al. (2010b) also described a milk production loss in multiparous cows with higher NEFA concentrations prepartum

(≥ 0.33 mEq/L within 2 wk prepartum). Timing of lipolysis in relation to calving appears to be important for interpretation of effects on milk yield. A milk loss in multiparous cows only among animals with NEFA concentration ≥ 0.5 mEq/L in the last week prepartum, and ≥ 0.7 mEq/L in the first week postpartum, has also been described by Chapinal et al. (2012a). However, in the same study, the authors also evaluated associations of NEFA concentrations in the first 2 wk postpartum with the first 4 weekly milk tests and found that cows above the established NEFA threshold produced more milk than those below the threshold, indicating that lipolysis may be critical to support higher milk yields (Chapinal et al., 2012a). Associations at the herd level between high postpartum NEFA and milk yield were found by Ospina et al. (2010c) but not others (Chapinal et al., 2012b). It is noteworthy to mention that cows with higher NEFA concentrations often show greater differences in FCM yields due to a higher concentration of preformed FA integration into milk fat, and lower contribution of de novo FA in milk fat (Mann et al., 2015, 2016c).

From these data, it appears that low prepartum body condition was most often associated with negative consequences on milk production in the next lactation. Rapid and excessive lipolysis in the immediate peripartum period, at least in multiparous cows, may also be detrimental to achieving a high milk production. Heifers may mobilize AT due to different reasons than cows. Regardless, the ability to mobilize AT is an integral part of supporting the energetic cost of milk production, particularly in high-producing cows. This was well demonstrated by Harrison et al. (1990) in high- versus average-producing cows and by an increased lipolytic rate of AT biopsies in high versus low genetic merit cows (Rocco and McNamara, 2013).

THE EFFECT OF ADIPOSE TISSUE IN THE TRANSITION PERIOD ON REPRODUCTION

The relationship of AT mass in the transition period with reproductive outcomes in early lactation has been well described. Low BCS has been associated with increased risk of developing uterine disease. In a study on 234 German Holstein cows, those with a BCS < 3.0 at calving had an almost 3 times higher odds of developing endometritis (Hoedemaker et al., 2009). Similarly, in a large number of North American cows, those with BCS ≤ 2.75 at parturition had approximately 2 times greater odds to develop cytological endometritis (Dubuc et al., 2010).

Low BCS also appears to delay the onset of estrus postpartum. A BCS < 3.0 at calving and at AI as well as a greater BCS loss postpartum (≥ 1 unit) were as-

sociated with delayed onset of estrus (Santos et al., 2009), and the proportion of anovulatory cows at AI was greater than 30% in cows with BCS < 2.75 . Pastured cows fed supplemental silage and grain and with a calving BCS of 4 versus 5 on a 10-point scale (corresponding to 2.78 vs. 3.10 on a 5-point scale; Roche et al., 2004) had a 9-d longer anovulatory interval postpartum (Roche et al., 2015).

Cardoso et al. (2013), using combined data from 7 different transition-cow feeding studies, classified dry cows 4 wk before calving and at 1 wk postpartum into thin (BCS < 3.0), moderate (BCS 3–3.75), and fat (BCS > 3.75) cows. When scored 4 wk prepartum, thin cows had a 91-d longer median time to pregnancy in the following lactation compared with fat cows. When scored at 1 wk postpartum, cows classified as thin had a 22-d increased median time to pregnancy compared with moderate BCS cows (Cardoso et al., 2013). Using a meta-analysis, López-Gatius et al. (2003) determined that low BCS (< 2.5) at parturition and at first insemination was associated with decreased pregnancy rate and increased days open. Cows with BCS ≤ 2.75 at 70 DIM and with clinical mastitis before insemination had a 2-fold increased risk of pregnancy loss (Hernandez et al., 2012).

These data linking low BCS in the peripartum period to poor reproductive success and an increased risk for reproductive tract disease may suggest that underconditioning of cattle at this time is the most undesirable situation, and that condition of these cows should be improved prepartum by feeding a higher energy diet to build up condition. However, in the same combined analysis performed at the University of Illinois described above, Cardoso et al. (2013) showed that overfeeding energy in the close-up period was associated with a 10-d increase in time to pregnancy and greater BCS loss postpartum. On the other hand, when diet was not manipulated and BCS change in cows was simply recorded in an observational study design, cows that gained BCS from -21 to 21 DIM had $2\times$ and $3\times$ greater chance of pregnancy or AI than cows that maintained or lost BCS between -21 and 21 DIM (Barletta et al., 2017). It can be hypothesized that in the latter study, BCS differences were due to intake differences (and possibly underlying health disorders) within a single group, which would represent a different situation than controlling body condition gain in an apparently healthy population by feeding a controlled energy diet versus overfeeding energy in the close-up period. Further, increasing energy density of the diets to attempt a BCS increase in the transition period would not have beneficial effects on reproduction. Unfortunately, intake data were not available, and prepartum and postpartum periods were not assessed

independently of each other in Barletta et al., (2017). Cows dried off at an undesirably high BCS of 4.08 and that subsequently lost ≥ 0.75 BCS units during the dry period had increased risk of uterine disease postpartum and reduced likelihood of pregnancy to first and second AI (Chebel et al., 2018). Similarly, overfeeding energy prepartum to achieve overconditioning prepartum [~ 60 MJ/d of NEL in excess of predicted requirements (60 MJ ~ 14.3 Mcal)] was associated with a greater risk of fatty liver, higher postpartum NEFA concentrations, and a 16-d longer interval to first ovulation (Rukkwamsuk et al., 1999). Cows with higher liver TAG (> 50 mg/g of wet weight) had increased interval to first heat and the probability of pregnancy was decreased by 30% (Jorritsma et al., 2000).

Associations of AT mobilization, expressed as NEFA concentrations in blood, and reproductive outcomes have been explored in several studies. The relationship between elevated prepartum NEFA concentrations and retained placenta or metritis risk (Dubuc et al., 2010; Ospina et al., 2010a; Chapinal et al., 2011) is likely explained by a decrease in energy intake resulting in higher NEFA in cows at risk for developing metritis postpartum (Huzzey et al., 2007). Higher postpartum NEFA concentrations (≥ 0.72 mEq/L) were found to be detrimental for pregnancy at first AI (Garverick et al., 2013), and reduced the risk of pregnancy by 16% within 70 d after the voluntary waiting period (Ospina et al., 2010b). Cows that lost BCS between -21 and 21 DIM and with higher postpartum NEFA concentrations compared with their herd mates had a decreased proportion of cows cycling at 50 DIM (Barletta et al., 2017). In vitro, increased concentrations of NEFA were found to be detrimental to oocyte and embryo development (Leroy et al., 2005; Van Hoeck et al., 2011; Jordaens et al., 2020). Cows that lost body condition early postpartum experienced greater pregnancy loss than cows that maintained condition (Middleton et al., 2019) and had reduced embryo quality (Carvalho et al., 2014).

THE IMPORTANCE OF TRANSITION PERIOD ADIPOSE TISSUE ON DISEASE RISK

The effects of AT mass on disease risk have been the subject of recent reviews (Roche et al., 2009, 2013). Due to the importance of AT in the adaptation to lactation explained above, and the direct link of AT mobilization with energy balance, it is not surprising that associations of either over- or underconditioning have been investigated for most disease processes common to the transition period.

Overconditioning of dry cows has long been identified as a risk factor for postpartum ketosis and fatty liver

(Morrow, 1976; Gillund et al., 2001; Bobe et al., 2004). Cows with a BCS ≥ 4 at dry off had an at least $2\times$ greater incidence of postpartum hyperketonemia and had over $8\times$ greater odds to develop clinical ketosis than cows with BCS ≤ 3 (Vanholder et al., 2015; Rathbun et al., 2017). Fatty liver risk is highest in obese cows with BCS ≥ 4 at calving, and it is questionable if thin cows can develop fatty liver due to the limited ability to mobilize AT (Bobe et al., 2004). Fatty liver is associated with lower expression of function-associated surface markers of blood-derived neutrophils and lower ex vivo-stimulated reactive oxygen species production and ability to lyse target cells (Zerbe et al., 2000).

High BCS on an individual and herd level is a known risk factor for displaced abomasum (DA) with a $2.4\times$ greater risk for cows to develop a DA with every 1-unit increase in BCS (Cameron et al., 1998). Greater lipolysis in the peripartum period expressed by NEFA concentrations ≥ 0.5 mEq/L at 1 wk prepartum or ≥ 1.0 mEq/L at 1 wk postpartum were associated with greater than twice the odds of developing a DA (Chapinal et al., 2011). Due to greater window of sampling days relative to parturition, cut-offs varied slightly in a study by Ospina et al. (2010a; ≥ 0.27 and 0.72 mEq/L for pre- and postpartum, respectively), but similar associations were found.

Excessive and low body condition in grazing cows have both been associated with increased risk of milk fever (Roche and Berry, 2006), but, to the author's knowledge, the association of AT and milk fever risk has not been well explored in confined dairy systems. However, the association between body condition and lameness has been well documented. In a multiyear study of one herd analyzing 79,565 cow-weeks at risk, BCS < 2 was associated with greater risk of lameness events (Randall et al., 2015). This association has also been found by other researchers (Espejo et al., 2006; Hoedemaker et al., 2009; Green et al., 2014) and a possible etiology with reduced shock absorbance due to a decreased digital adipose cushion thickness in thin cows is postulated (Bicalho and Oikonomou, 2013; Griffiths et al., 2020) and may lead to increased risk of sole hemorrhages (Wilhelm et al., 2017).

Associations of BCS on udder health have not been evaluated to the same degree as the aforementioned diseases. In one study, older and heavier cows with greater mobilization of BW had higher SCC scores (Berry et al., 2007).

FETAL PROGRAMMING OF ADIPOSE TISSUE

Fetal adiposity can be programmed in utero, likely through altered circulating concentrations of endocrine mediators such as IGF-I and leptin (Bispham et al.,

2003). Indeed, in sheep, nutrient restriction before d 80 of pregnancy led to greater fetal AT deposition, regardless of the energy intake of their dams in late gestation (Bispham et al., 2003). Restriction to 50% of nutrient requirements until d 80 of pregnancy in sheep was associated with increased postnatal weight gain and adiposity (Ford et al., 2007). Placental insufficiency induced by heat stress in pregnant ewes from d 39 to 96 of pregnancy was associated with lower birth BW and decreased AT mobilization of lambs in response to adrenaline and a greater reliance on glucose to meet metabolic demands (Chen et al., 2010).

In cattle, most of the available work of fetal programming of AT has been performed in beef cattle. Severe nutrient restriction to 65% of requirements in the first 82 d of pregnancy, followed by feeding to requirements, was associated with higher NEFA concentrations in offspring of Spanish beef cattle at 13 mo of age (Noya et al., 2019). Feeding a higher energy density diet to late gestation Angus-cross cows was associated with greater intramuscular fat content of the progeny at slaughter (Radunz et al., 2012). Likewise, in humans, increased maternal food intake from mid-to-late gestation led to increased fetal AT at term (Symonds et al., 2012). It should be noted that the aforementioned dietary interventions are experimental and of severe magnitude, and likely of little importance to well-managed dairy cattle today.

Despite a growing interest in epigenetic differences, potential fetal programming effects on white AT methylation patterns have not been well studied to date due to high interanimal variability and species-specific differences (Symonds et al., 2012). Moreover, the direct influence of the dam's AT mass, rather than overall energy balance, on fetal AT programming of dairy cattle has not been investigated. In summary, the effects of feeding strategies of the pregnant cow and environmental conditions on fetal programming of AT require further investigation in future studies.

MODULATION OF INFLAMMATION

Rapid mobilization of AT releases adipokines and cytokines into circulation (Häussler et al., 2022) and appears to promote immune cell infiltration and even adipocyte death. Newman et al. (2019) and De Koster et al. (2018) showed that rapid postpartum BCS loss is associated with greater numbers of AT macrophages. It should be noted that immune cell infiltration varies by AT depot and is higher in the more difficult to sample visceral depots (Aker et al., 2012). Based on our current understanding, it is likely that these immune cells that increase due to local AT remodeling in cows with body condition loss do not contribute to systemic

inflammation. This is supported by the lack of inflammatory phenotype of macrophages that infiltrate AT after feed restriction in late-lactation cows (Contreras et al., 2016). However, *TNFA* gene expression increases in subcutaneous AT of dairy cows postpartum (Vailati Riboni et al., 2015; Mann et al., 2016b), possibly due to the increased number of immune cells at this time.

To further investigate the association of inflammation and lipolysis, a series of experiments were undertaken with recombinant TNF- α administrations into the subcutaneous AT depot. Initial work in cows 70 to 98 DIM showed that subcutaneous daily injections of TNF- α at 2.5 μ g/kg of BW increased lipolysis (Kushibiki et al., 2003). Subsequently, daily injections of TNF- α did not lead to increased NEFA concentrations, indicating a lack of AT lipolysis response in late-lactation cows, but did alter hepatic metabolism and hepatic triglycerides (Bradford et al., 2009). In 2 subsequent studies, the same group found that lipolysis was not increased due to daily TNF- α injections in fresh cows (Yuan et al., 2013). Additionally, AT of late-lactation cows responded with upregulation of anti-inflammatory cytokine gene expression to recombinant TNF- α and lipolysis was not increased (Martel et al., 2014).

The potential relationship of inflammation and AT mobilization warrants a closer look at the effect of anti-inflammatory interventions on AT modulation of the transition cow. It could be hypothesized that if inflammation increases lipolysis, quenching of inflammatory mediators would dampen the lipolytic activity.

EFFECTS OF NONSTEROIDAL ANTI-INFLAMMATORY TREATMENT ON LIPOLYSIS

In a prepartum 4-d feed restriction model, Pascottini et al. (2019) expected the induced lipolysis to lead to an increase in circulating proinflammatory cytokines and treated half of the 30 feed-restricted cows with 0.5 mg/kg of BW meloxicam subcutaneously. Despite substantial AT mobilization in both groups, systemic inflammation was not observed compared with a group fed normally, and NEFA concentrations did not differ between treated and untreated cows (Pascottini et al., 2019).

Postpartum, Schwartz et al. (2009) treated cows with 2.2 mg/kg of BW of flunixin meglumine for 3 d compared with an untreated control. Treated cows tended to have higher NEFA concentrations after treatments ended, consumed less feed during and after treatment, and showed lower milk production (Schwartz et al., 2009), indicating that nonsteroidal anti-inflammatory (NSAID) treatment with flunixin meglumine was detrimental to early-lactation metabolic adaptation. Farney et al. (2013) administered sodium salicylate in

water from 1 to 7 DIM at 1.95 g/L and found no effect on NEFA during treatment, but NEFA concentrations increased in the treated group after treatment ended, possibly caused by a rebound inflammatory response. In a similar study where cows were treated with 2.3 g/L of sodium salicylate in drinking water for 7 d postpartum, no differences in NEFA concentrations were shown during treatment (Montgomery et al., 2019). Following the same thought that NSAID administration may modulate AT lipolysis, Kovacevic et al. (2019) treated cows with 3 mg/kg of BW i.m. of ketoprofen 1 to 3 DIM and found that treated cows had lower serum NEFA concentrations and decreased inflammatory cytokine concentrations in the wk 2 and 3 postpartum; however, no data on intakes or milk production were presented. Carpenter et al. (2016) treated cows with either a bolus of 1 mg/kg of BW of meloxicam or 3 daily drenches of 185 g/kg of BW of sodium salicylate, and found no difference in NEFA between these groups and control cows. It is important to note that NEFA concentrations alone may not accurately reflect whole-body lipolysis due to the different fates of FA upon release into circulation as described above.

Using samples from the Montgomery study, Takiya et al. (2019) investigated inflammation in AT samples from treated and control cows with a proteomic approach. They found that NSAID treatment increased both the pro- and anti-inflammatory signaling pathways of AT. Interestingly, AT gene expression of several proinflammatory cytokines also increased in response to 1.4 mg/kg of BW of carprofen on d 1, 3, and 5 post-calving in another study (Vailati Riboni et al., 2015).

From these studies, it appears that AT is responsive to anti-inflammatory treatments, although the increase in proinflammatory response and NEFA concentrations during or after treatments may be counter-intuitive. Observed differences may be caused by an intake and DMI response to treatment, rather than a direct effect on AT, and the direct effect of NSAID on AT is difficult to dissect, particularly in the postpartum period. The biological significance of the treatment-associated changes seen thus far and that are specific to AT modulation should be investigated in future NSAID treatment trials carried out in the transition period.

MANAGEMENT STRATEGIES TO OPTIMIZE ADIPOSE TISSUE FUNCTION

Housing and Management

Housing and management factors influence AT lipolysis. Longer calving intervals are related to greater body condition at calving and associated with greater early-lactation BCS loss (Middleton et al., 2019). A

longer dry period (8 wk) was associated with greater BCS loss after calving than a 4-wk dry period (André O'Hara et al., 2018). Cows with a 90-d dry period had greater NEFA concentrations than those with 28- or 56-d dry periods, although BCS at dry off already differed between groups (Weber et al., 2015). This is relevant because BCS at dry off is directly related to feeding behavior and feeding time in dry cows, and it is recommended to avoid overconditioning at dry off to limit BCS loss prepartum (Daros et al., 2021).

Substantial forced exercise prepartum (2×45 min walking) induced prepartum lipolysis in overconditioned cows and may have improved lipid utilization before calving (Goselink et al., 2020). Milking frequency reduction and the associated drop in milk production were associated with a decrease in NEFA concentrations and a reduced BCS loss (Stelwagen et al., 2013). McNamara et al. (2008) compared $1\times$ versus $2\times$ and $3\times$ milking in the first 4 wk postpartum and found that cows in the $1\times$ group did not lose BCS and had lower NEFA concentrations than cows in the other 2 groups, but lower milk yield in $1\times$ cows was still present at 15 wk. Findings, including long-lasting negative effects on milk yield in the $1\times$ group, were similar for cows with reduced milking frequency on pasture (Phyn et al., 2014). Moallem et al. (2019) compared $2\times$ with $3\times$ milking in the first 30 DIM and found improvements in energy balance without lasting negative effects on milk production when cows were milked $2\times$. Increase in milking frequency from $3\times$ to $6\times$ in the first 21 DIM has been investigated by others (Shoshani et al., 2017), with long-lasting increases in milk production in cows milked $6\times$. Effects on BCS were parity dependent with a lower BCS in the $6\times$ group in second-lactation cows only. These findings underline the importance of AT lipolysis in support of the early-lactation energetic needs of milk production, which appear to be altered to the greatest degree when cows are milked only once daily compared with higher frequency of milking. In addition to factors that can be altered by on-farm management such as those presented above, BCS and change in BCS have a heritable component (Roche et al., 2009).

Heat Stress and Heat Abatement

Heat stress during the transition period can influence the degree of lipolysis. Heat stress normally reduces lipolysis in different agricultural species, including cows (Belhadj Slimen et al., 2016). However, the effect of heat exposure might be different in transition cows. Heifers calving in summer months without heat abatement had higher and more prolonged increases in NEFA concentrations postpartum compared with those calving in the winter (Turk et al., 2015). This difference

in transition animals may be caused by a lower DMI in combination with the particular metabolic situation of early lactation.

Given the known negative effects of heat stress, heat abatement is often used in dry- and fresh-cow groups. Cows cooled prepartum had greater milk production and only approximately half the BCS loss postpartum compared with heat-stressed cows, possibly due to the higher DMI (Karimi et al., 2015). In 2 other studies, cows cooled in the dry period had higher NEFA concentrations postpartum and greater BCS loss (do Amaral et al., 2011; Tao et al., 2012). This difference may be explained by the significantly higher milk production in cows that were cooled before calving and their greater energy needs that were not met by higher intake (Tao et al., 2019) or by a direct effect of heat stress on suppression of lipolysis. Indeed, cows that were cooled postpartum had minimally higher NEFA concentrations (+0.04 mEq/L), and higher DMI and milk production compared with heat-stressed cows (Safa et al., 2019). Therefore, the effect of heat stress during the transition period appears to depend on the cow's DMI and milk response in relation to the applied cooling.

Nutrition and Diet Supplementation

The effect of dietary strategies; nutritional supplementation and treatments with insulin, glucose, glucocorticoids; and glucogenic precursors on lipolysis modulation have been recently reviewed in detail (Contreras et al., 2018). Only selected studies will be highlighted here that provide additional key points or important updates to the existing literature.

Prepartum Energy Intake

Restricting the energy density of dry cow diets to meet but not exceed energy requirements in an ad libitum feeding system has been associated with lower NEFA concentration postpartum (Mann et al., 2015; Richards et al., 2020). This difference could have been caused by the greater AT deposition in overfed cows due to overconsumption of energy, although only small (Richards et al., 2020) or no differences (Mann et al., 2015) in BCS were found, and it is likely that AT deposition was mostly in the visceral AT (Drackley et al., 2014), which was not measured in either study. This difference has not been shown to be associated with a change in AT insulin resistance (Ji et al., 2012; Mann et al., 2016d). However, postpartum glucose concentrations in cows overfed energy prepartum were lower than those fed a controlled energy diet throughout the dry period (Mann et al., 2016d). Lower glucose concentra-

tions may have driven the difference in lower circulating concentrations of insulin, allowing for the greater degree of NEFA release by lipolysis observed in this study, and a significantly increased concentration of glucagon in the overfed group early postpartum supports this (Mann et al., 2016d).

Supplemental Fat and Fatty Acid Sources

Feeding of supplemental fat is a strategy to increase energy intake, and effects of fat feeding on AT metabolism have been studied for several decades (Vernon, 1981). The profile of FA in the supplemental fat is important, as feeding palmitic acid increased the loss of BW and BCS compared with a control diet in the fresh period and, to a smaller degree, during peak lactation (de Souza and Lock, 2019). Feeding an increasing amount of oleic acid, on the other hand, decreased BCS loss, possibly due to an increased circulating concentration of insulin and DMI despite an increase in milk energy output (de Souza et al., 2021). The difference in FA source included in transition diets on BW loss has been evaluated using a meta-regression (Rodney et al., 2015), and overall little effect on BW changes during the period were found. However, heterogeneity in the data was present; therefore, interpretation was limited (Rodney et al., 2015).

Further underlining the differential effects of FA on AT of the transition cow are a series of experiments on CLA, known to be potent inhibitors of milk fat synthesis (Harvatine et al., 2018). Feeding rumen-protected CLA for 3 wk before and after calving decreased milk fat and NEFA concentrations, downregulated AT gene expression of lipolytic enzymes, and decreased BCS loss (Dirandeh et al., 2021). Authors explained the difference in AT mobilization by a decrease in the energy needed to produce milk fat in CLA-supplemented cows (Dirandeh et al., 2021). A decrease in milk fat content and lower BCS loss was also found by others that fed CLA in the postpartum period (Galamb et al., 2017; Qin et al., 2018), but effects on decreasing AT mobilization were not observed by all (von Soosten et al., 2011). When supplemented for the last 18 d prepartum only, but not after calving, CLA feeding did not have an effect on milk fat, but decreased NEFA in wk 1 postpartum (Oliveira et al., 2018). In summary, AT lipolysis was modulated by CLA at the cost of reducing the fat component of milk when fed during lactation, and generally a decrease in AT loss was found with CLA feeding. However, feeding of CLA to modulate AT loss at the cost of milk fat production may not be a viable strategy for producers paid on a milk component basis (Oliveira et al., 2018).

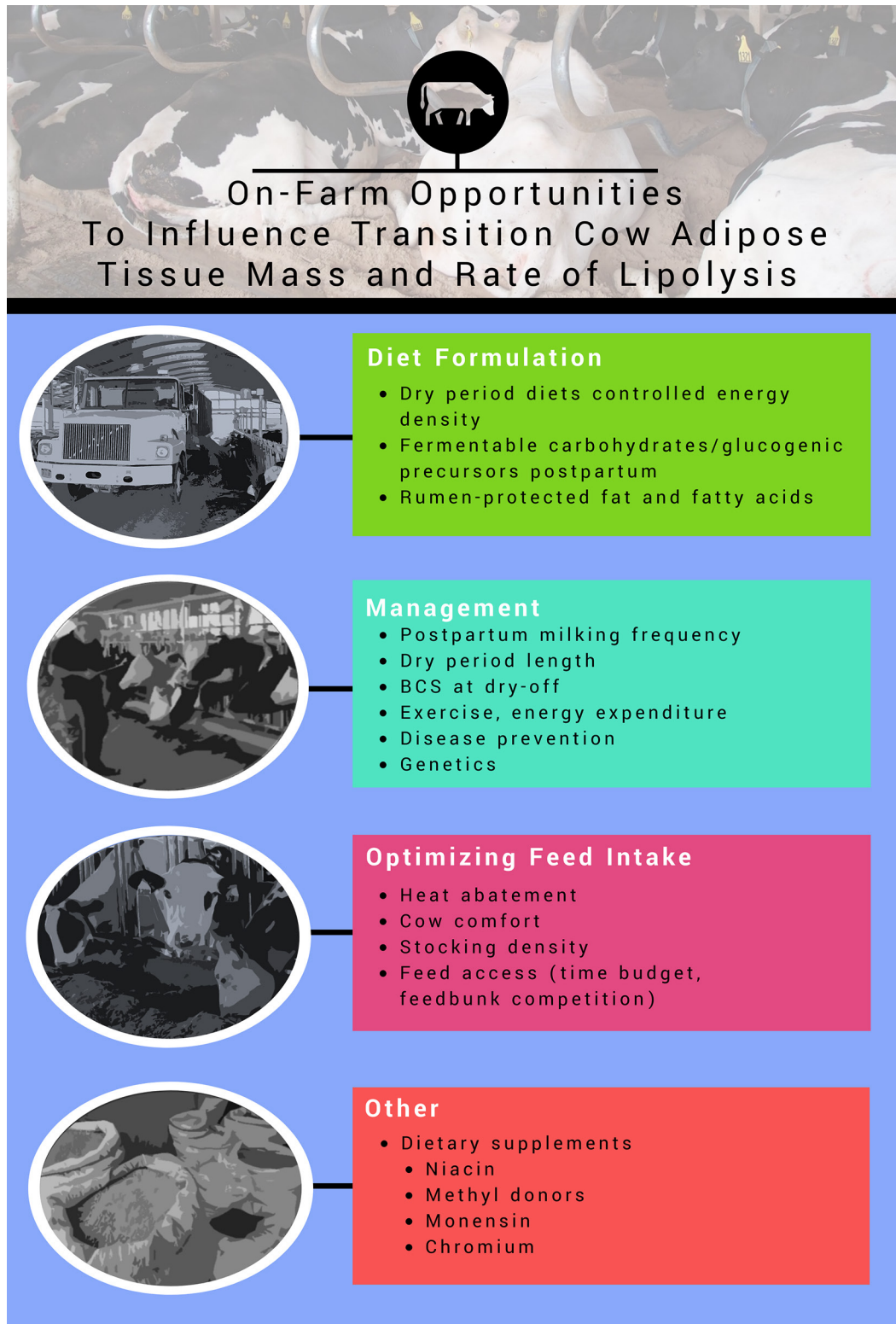


Figure 1. Summary of on-farm opportunities to influence adipose tissue mass and rate of lipolysis of transition cows. See text for details.

KNOWLEDGE GAPS AND FUTURE OPPORTUNITIES

Several knowledge gaps emerged from the review of the current literature on the role of AT in the transition period. First, better ways to monitor, the visceral AT mass in particular, are needed that can be used both in research and for on-farm management (Figure 1). Given that BCS does not correspond well to total AT mass (Drackley et al., 2014), and that visceral AT has a higher lipolytic activity than subcutaneous AT (Ruda et al., 2019), the usefulness of BFT or BCS for these purposes may be questioned. Visceral AT mass exceeds subcutaneous AT by 2 to 3 times (Szura et al., 2020). Estimation of total AT mass, rather than BCS or BFT measurements, would provide more accurate data on the effects of management or feeding interventions on AT, and subsequently of the role of total AT mass on outcomes of interest such as health, productivity, and reproductive success. This is particularly important as subcutaneous AT mass appears to be more resistant to change in the transition period, whereas visceral AT has a greater rate of gain and loss in the pre- and postpartum periods, respectively. Additionally, visceral AT loss is related more closely to the previous rate of gain than to calculations of energy balance (Szura et al., 2020). Therefore, only focusing on subcutaneous AT will underestimate the dynamic changes of AT in the transition period. Definition of an optimal AT mass at dry off and calving, and acceptable loss of AT mass, may be redefined once these visceral measurements can be used widely on farms.

Second, the role of fetal programming of AT on future health and productivity of the dairy calves is largely uncharted territory. Adipose tissue is central to the dairy cow's ability to respond to times of energy deficit. The effects of fetal programming on the calf's early life and later AT biology are only beginning to emerge. The question of how the dam's metabolic status and exposure to stressors such as heat stress during pregnancy affects her offspring is not clear to date and represents a knowledge gap. When a better understanding of fetal programming of AT exists, opportunities may present themselves to integrate this knowledge into transition-cow management decision making.

Last, the field of transition-cow biology is making advances in the understanding of systemic inflammation on metabolic adaptation; however, the role of AT as an instigator or innocent bystander of these changes is not fully understood. Whether immune cell infiltration and greater expression of inflammatory cytokines in AT postpartum are normal and desirable physiological responses to the required lipolysis of early lactation, or if AT inflammation is an active contributor to failed homeorhetic adaptation with associated negative health

and productivity outcomes, needs to be investigated. An understanding of this question will be required to plan and evaluate anti-inflammatory treatments and their effects on AT modulation of the transition cow.

CONCLUSIONS

The importance of the dairy cow AT as an efficient caloric reservoir in the adaptation to lactation is undebatable. Underconditioning of cows and rapid and excessive mobilization of AT have been linked with poor performance and negative health events as well as reduced reproductive success. Management and environmental factors such as milking frequency, dry period length, and heat stress abatement can modulate AT in the transition period. Nutritional strategies and dietary supplementation are widely used and accessible strategies to modulate AT in the transition period, and should aim to minimize gain and loss of AT as much as possible. Fetal programming of AT in utero and future productivity and health, as well as the effects of anti-inflammatory treatments warrant further research. A key knowledge gap is the contribution of visceral AT to health and production outcomes, and the role of AT inflammation in normal transition-cow biology and pathophysiology. A simple, rapid, repeatable, and accurate method to describe visceral AT mass in addition to subcutaneous AT mass is needed.

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Mann: ANIMAL HEALTH SYMPOSIUM

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